EXP NO:2 Study of different type of Network cables

AIM:

Study of different types of network cables and under the procedure for making Ethernet straight - through and crossover cables.

Cable Type	Category	Max Data Transmission	Advantage / Disadvantage	Application
UTP	Category 3	10 bps	Advantage: • cheap cost • Easy to small install Disadvantage • More prone to EMI	10 Base T Ethernet Fast Ethernet Gigabit Ethernet Fast Ethernet Gigabit Ethernet.
	Category 5	up to 100 Mbps		
	Category 5e	1 Gbps		
STP	Category 8/8a	10 Gbps	Advantage: • shielded • Faster than UTP • Less susceptible to noise Disadvantage: • Expensive • Greater installation effort	Gigabit Ethernet (10G Ethernet (55m) widely used in data Gigabit Ethernet (10G Ethernet (100m)
SSTP	Category 7	10 Gbps		
Coaxial cable	RG-8 RG-59 RG-11	10-100 Mbps	Advantage: • High bandwith • Immune to interference • Versatile Disadvantage: • Disadvantage of cost • Size is bulky	Speed of signal is 500m High speed internet connection.
Fibre optics cable	Single mode Multi mode	100 Gbps	Advantage: • High speed • High bandwidth • High security Disadvantage: • Expensive • Required skilled installers	Maximum distance of fibre optics cable is around 100 miles

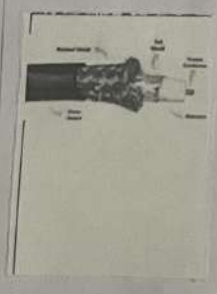
b) Make your own Ethernet cross over cable

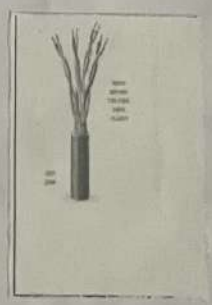
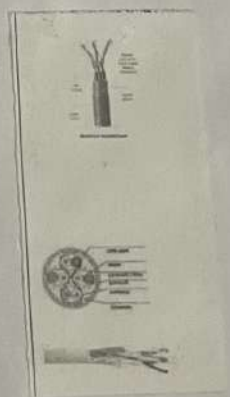
Tools and parts needed:

- Ethernet cabling. CAT5e is certified for gigabit.
- A crimping tool. this is an all-in-one networking tool shaped to push down the pins in the plug
- Two RJ45 plugs
- optional two plug shields.

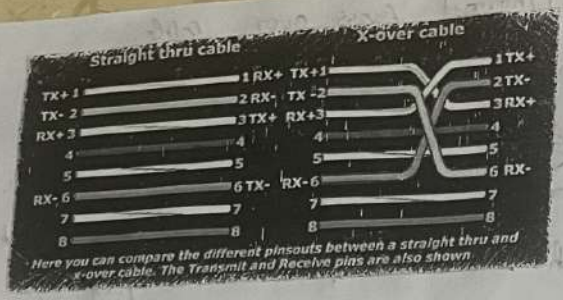
Procedure for Making Ethernet cable

1. Insert the cable boot
2. strip about 1.5 cm of the outer insulation
3. untwist the four wire pairs
4. Arrange the wires according to the required color code
5. Trim the wire evenly
6. Insert the wire fully into the RJ45 connector
7. Crimp the connector using the crimping tool
8. Repeat the same process for the other end
9. Test the cable using tester or by connecting two divices.

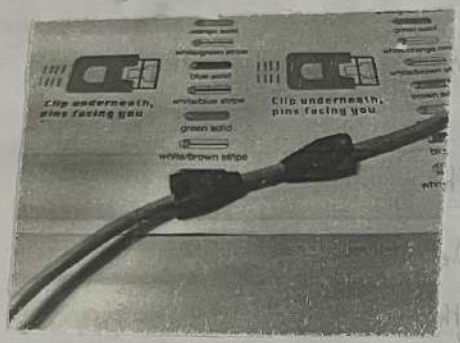


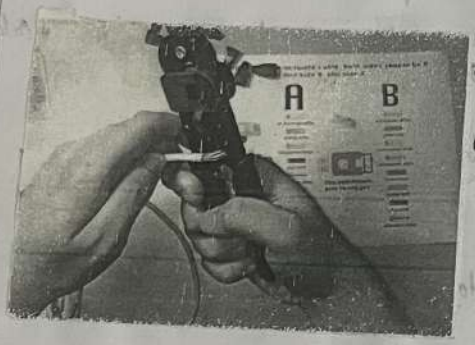
Network Cables



Difference between cross cable and straight cable



threading shields into cables



untangle the wire to form four twisted pairs



cable shielding from both the end.



Arrange the wire in order, trim any uneven ends insert them into RJ45 plug with the clip facing away, and gold pins toward you



the wire in the plug are inserted into the crimping tool and pushed down

RESULT:

Study of different types of network cables through cables and cross over is done successfully.